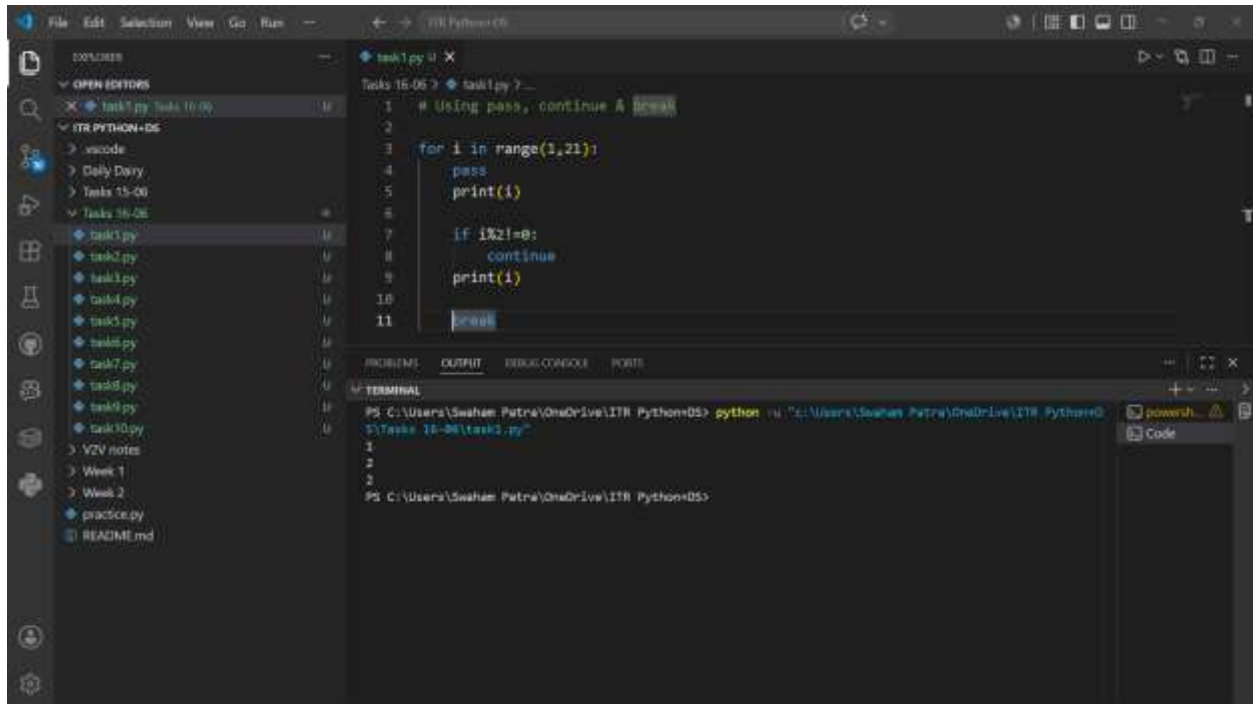


Swaham Pitambar Patra

Task 1:



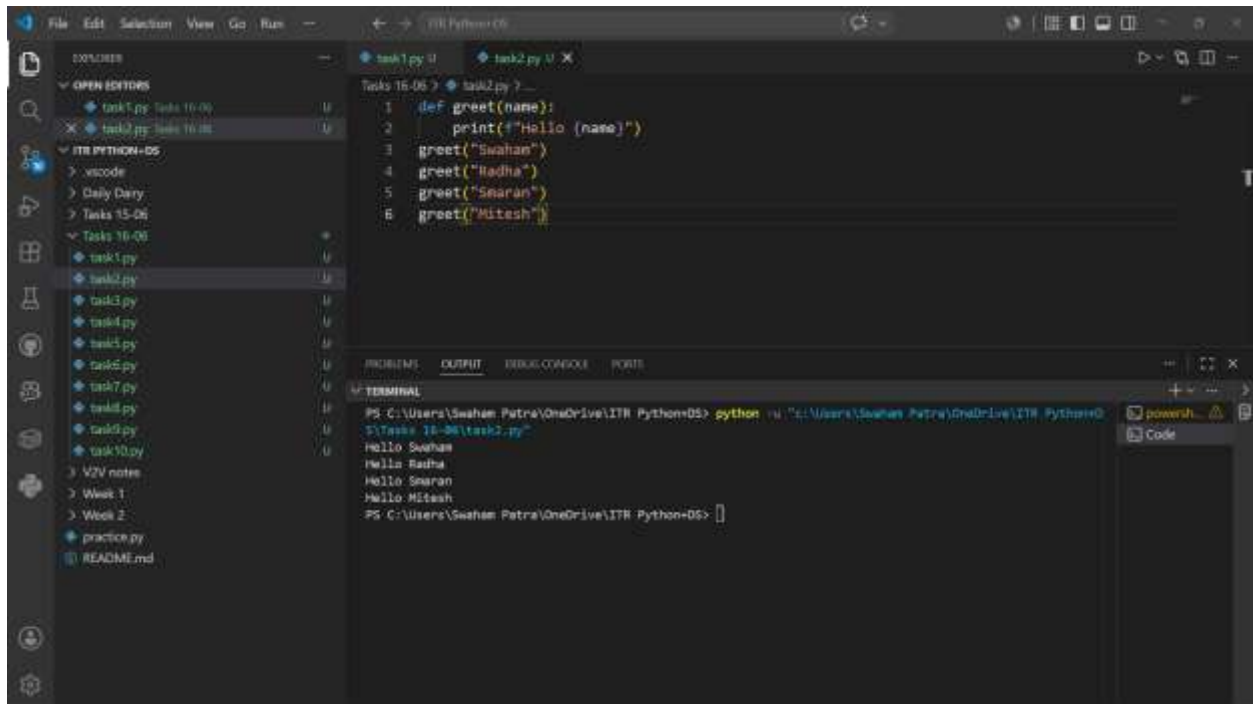
The screenshot shows the VS Code editor with a file named `task1.py` open. The code is as follows:

```
1 # Using pass, continue & break
2
3 for i in range(1,21):
4     pass
5     print(i)
6
7     if i%2!=0:
8         continue
9     print(i)
10
11 break
```

The terminal output shows the execution of the script:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python\OS> python -u "C:\Users\Swaham Patra\OneDrive\ITR Python\OS\Task1-16-06\task1.py"
1
2
2
PS C:\Users\Swaham Patra\OneDrive\ITR Python\OS>
```

Task 2:



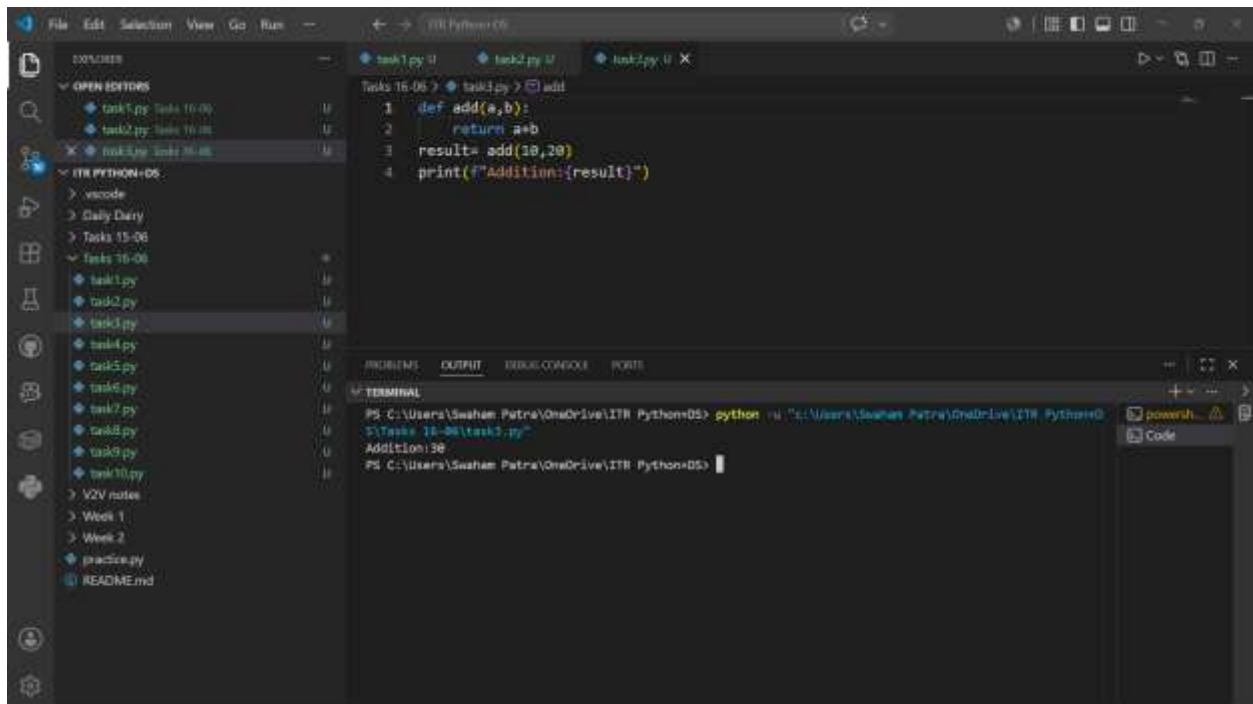
The screenshot shows the VS Code editor with a file named `task2.py` open. The code is as follows:

```
1 def greet(name):
2     print(f"Hallo {name}")
3     greet("Swahan")
4     greet("Radha")
5     greet("Snehan")
6     greet("Mitesh")
```

The terminal output shows the execution of the script:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python\OS> python -u "C:\Users\Swaham Patra\OneDrive\ITR Python\OS\Task1-16-06\task2.py"
Hallo Swahan
Hallo Radha
Hallo Snehan
Hallo Mitesh
PS C:\Users\Swaham Patra\OneDrive\ITR Python\OS>
```

Task 3:

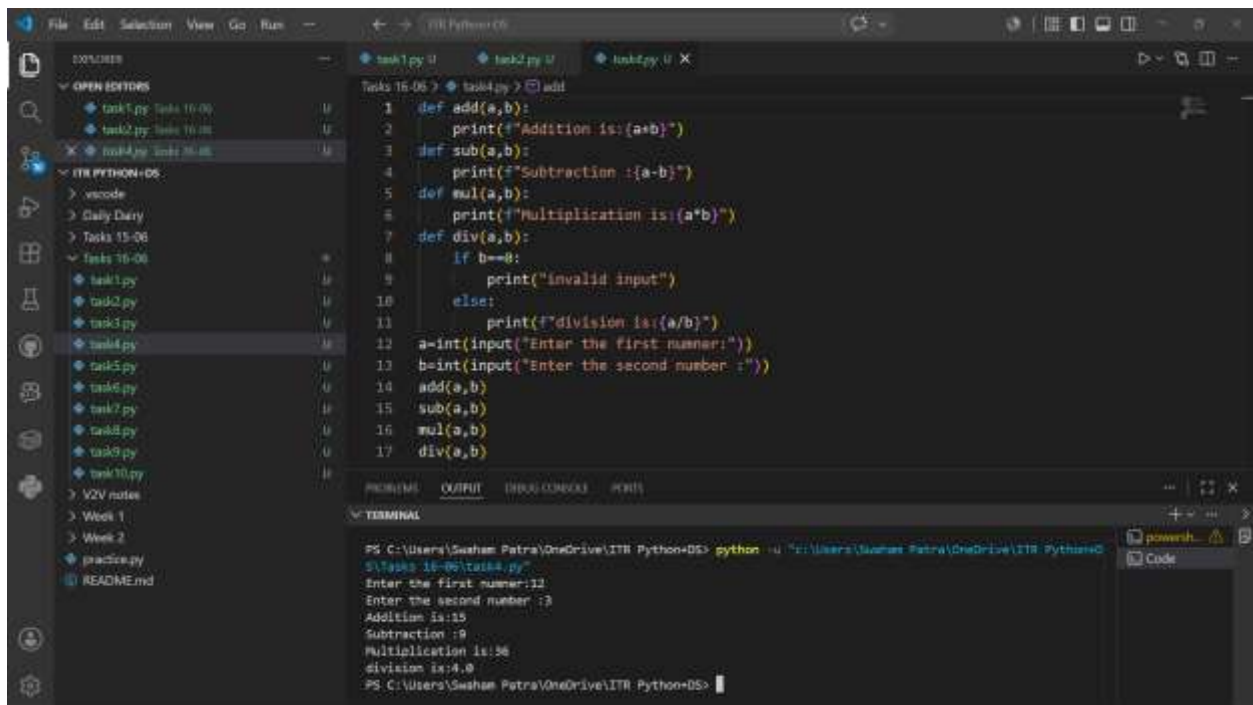


The screenshot shows the Visual Studio Code interface with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project named 'ITR Python DS' with a folder 'Tasks 15-06' containing files 'task1.py', 'task2.py', and 'task3.py'. The code editor displays the content of 'task3.py', which defines an 'add' function and calls it with 'add(10,20)'. The terminal shows the command 'python -u "C:\Users\Saaham Patra\OneDrive\ITR Python\DS\Tasks 15-06\task3.py"' and the output 'Addition:30'.

```
def add(a,b):
    return a+b
result= add(10,20)
print(f"Addition:{result}")
```

```
PS C:\Users\Saaham Patra\OneDrive\ITR Python\DS> python -u "C:\Users\Saaham Patra\OneDrive\ITR Python\DS\Tasks 15-06\task3.py"
Addition:30
PS C:\Users\Saaham Patra\OneDrive\ITR Python\DS>
```

Task 4:

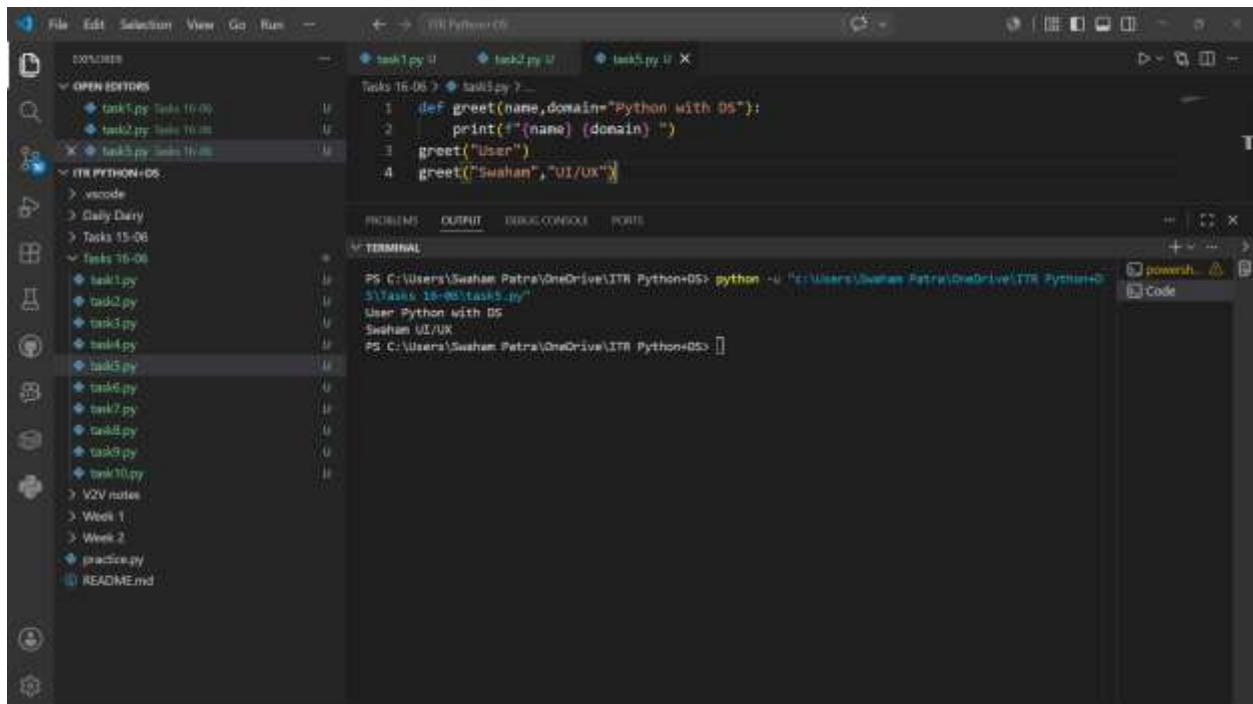


The screenshot shows the Visual Studio Code interface with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project named 'ITR Python DS' with a folder 'Tasks 15-06' containing files 'task1.py', 'task2.py', and 'task4.py'. The code editor displays the content of 'task4.py', which defines functions for addition, subtraction, multiplication, and division, and includes a main loop that takes user input and calls these functions. The terminal shows the command 'python -u "C:\Users\Saaham Patra\OneDrive\ITR Python\DS\Tasks 15-06\task4.py"' and the output of the program, including prompts for the first and second numbers and the results of the operations.

```
def add(a,b):
    print(f"Addition is:{a+b}")
def sub(a,b):
    print(f"Subtraction :{a-b}")
def mul(a,b):
    print(f"Multiplication is:{a*b}")
def div(a,b):
    if b==0:
        print("invalid input")
    else:
        print(f"division is:{a/b}")
a=int(input("Enter the first number:"))
b=int(input("Enter the second number :"))
add(a,b)
sub(a,b)
mul(a,b)
div(a,b)
```

```
PS C:\Users\Saaham Patra\OneDrive\ITR Python\DS> python -u "C:\Users\Saaham Patra\OneDrive\ITR Python\DS\Tasks 15-06\task4.py"
Enter the first number:12
Enter the second number :3
Addition is:15
Subtraction :9
Multiplication is:36
division is:4.0
PS C:\Users\Saaham Patra\OneDrive\ITR Python\DS>
```

Task 5:

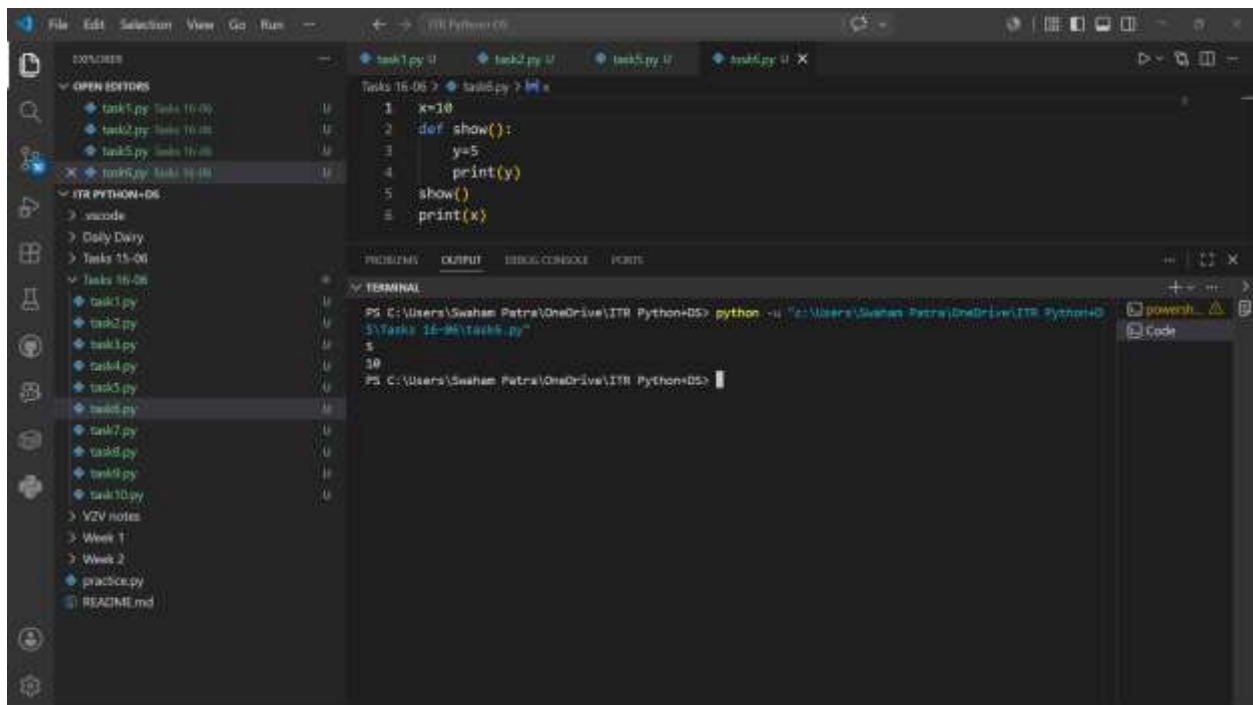


The screenshot shows the VS Code editor with the file explorer on the left. The 'TASKS' folder is expanded, showing 'task1.py' through 'task10.py'. The 'task5.py' file is open in the editor. The code defines a 'greet' function that takes 'name' and 'domain' as arguments. It prints a message and calls itself recursively with 'User' and 'Swaham UI/UX'.

```
1 def greet(name,domain="Python with DS"):
2     print(f"({name}) ({domain}) ")
3     greet("User")
4     greet("Swaham","UI/UX")
```

The terminal at the bottom shows the command 'python -u "C:\Users\Swaham Patra\OneDrive\ITR Python\DS\Tasks 15-06\task5.py"' and the output: 'User Python with DS' followed by 'Swaham UI/UX'.

Task 6:

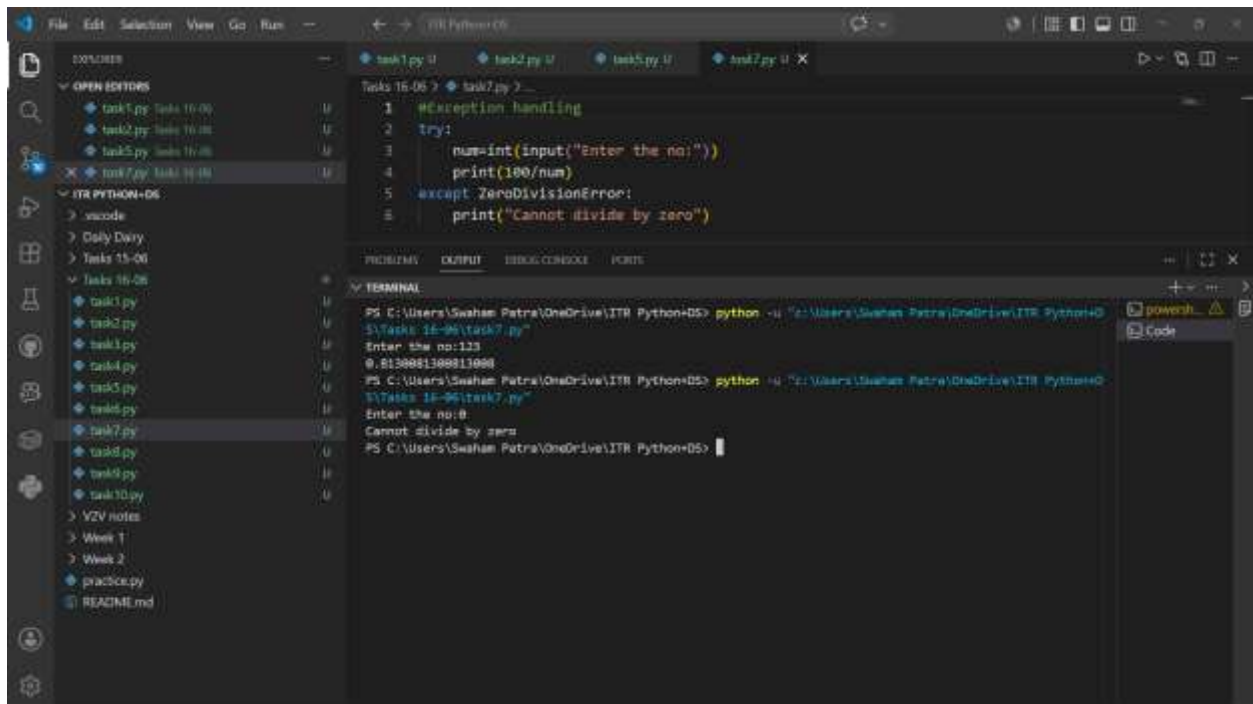


The screenshot shows the VS Code editor with the file explorer on the left. The 'TASKS' folder is expanded, showing 'task1.py' through 'task10.py'. The 'task6.py' file is open in the editor. The code defines a 'show' function that takes 'x' as an argument. It prints 'y' (which is 5) and calls itself recursively with 'x' (which is 10).

```
1 x=10
2 def show():
3     y=5
4     print(y)
5     show()
6     print(x)
```

The terminal at the bottom shows the command 'python -u "C:\Users\Swaham Patra\OneDrive\ITR Python\DS\Tasks 15-06\task6.py"' and the output: '5' followed by '10'.

Task 7:



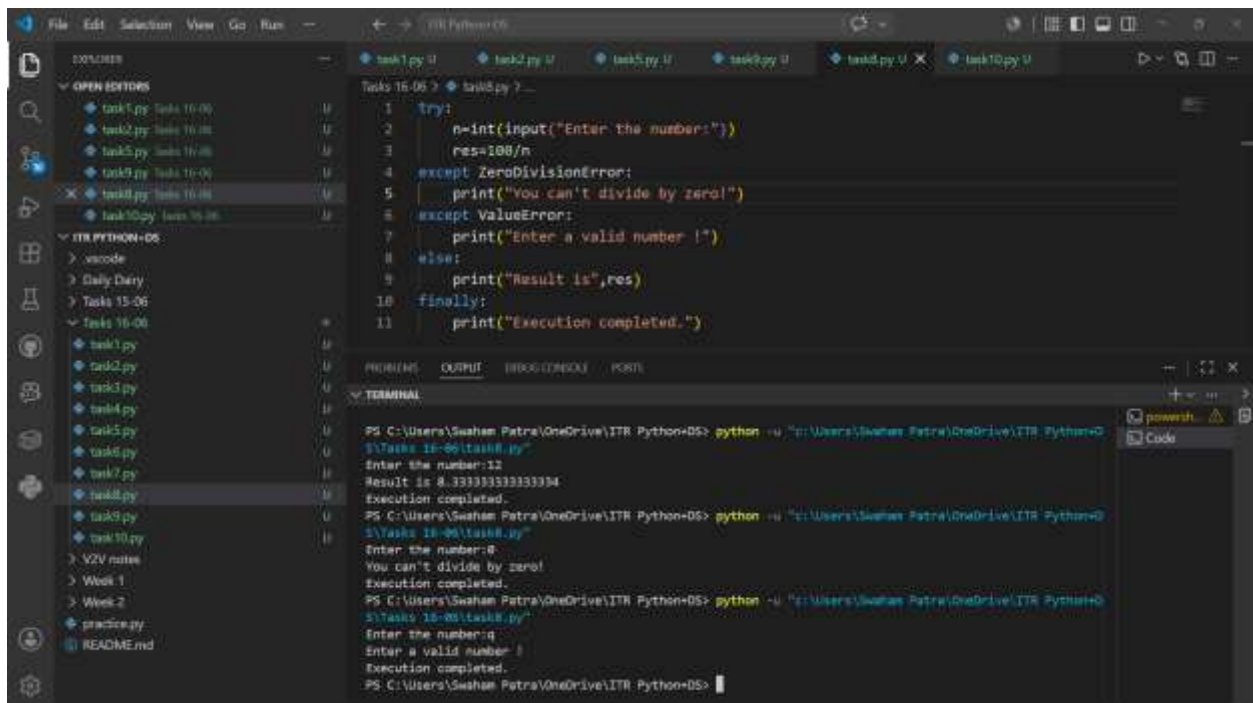
The screenshot shows the VS Code interface with the file explorer on the left displaying a project structure under 'ITR PYTHON-05'. The main editor shows 'task7.py' with the following code:

```
1 #Exception handling
2 try:
3     num=int(input("Enter the no:"))
4     print(100/num)
5 except ZeroDivisionError:
6     print("Cannot divide by zero")
```

The terminal at the bottom shows the execution of the script:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\05\Tasks 15-06\task7.py"
Enter the no:123
0.8130081300813008
PS C:\Users\Swaham Patra\OneDrive\ITR Python> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\05\Tasks 15-06\task7.py"
Enter the no:0
Cannot divide by zero
PS C:\Users\Swaham Patra\OneDrive\ITR Python>
```

Task 8:



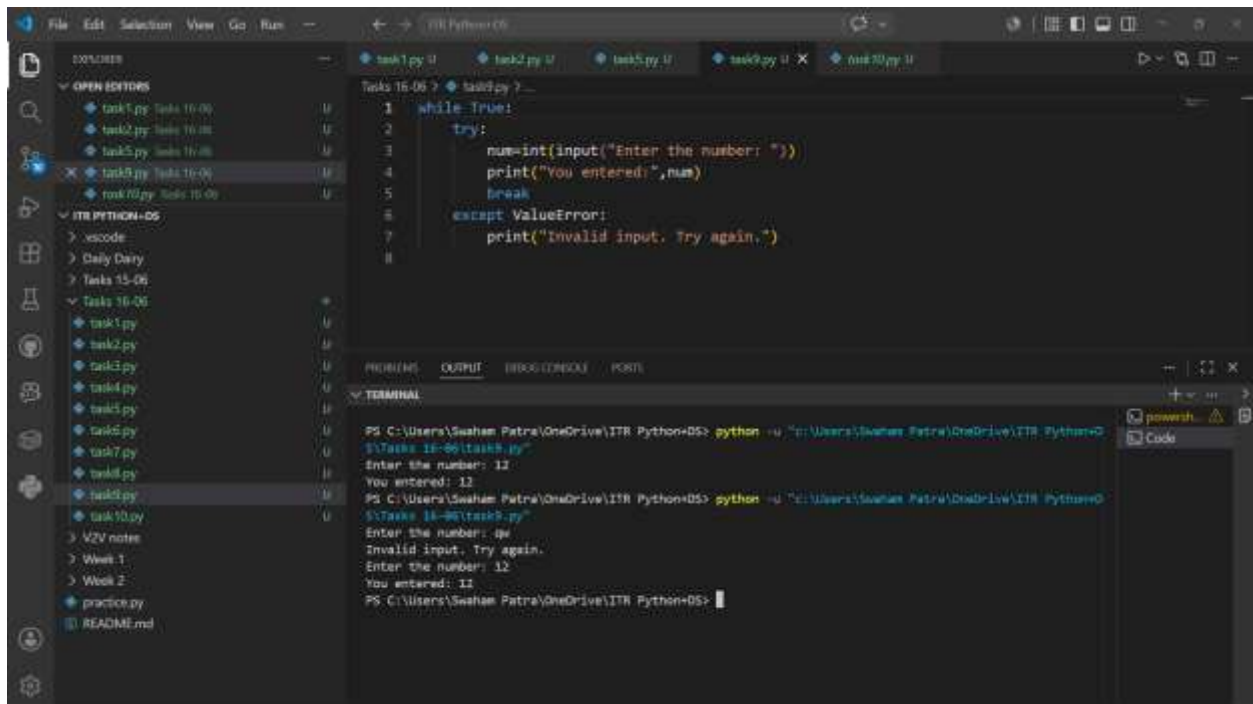
The screenshot shows the VS Code interface with the file explorer on the left displaying a project structure under 'ITR PYTHON-05'. The main editor shows 'task8.py' with the following code:

```
1 try:
2     n=int(input("Enter the number:"))
3     res=100/n
4 except ZeroDivisionError:
5     print("You can't divide by zero!")
6 except ValueError:
7     print("Enter a valid number !")
8 else:
9     print("Result is",res)
10 finally:
11     print("Execution completed.")
```

The terminal at the bottom shows the execution of the script for three different inputs:

```
PS C:\Users\Swaham Patra\OneDrive\ITR Python> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\05\Tasks 15-06\task8.py"
Enter the number:12
Result is 8.333333333333334
Execution completed.
PS C:\Users\Swaham Patra\OneDrive\ITR Python> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\05\Tasks 15-06\task8.py"
Enter the number:0
You can't divide by zero!
Execution completed.
PS C:\Users\Swaham Patra\OneDrive\ITR Python> python -u "c:\Users\Swaham Patra\OneDrive\ITR Python\05\Tasks 15-06\task8.py"
Enter the number:q
Enter a valid number !
Execution completed.
PS C:\Users\Swaham Patra\OneDrive\ITR Python>
```

Task 9:



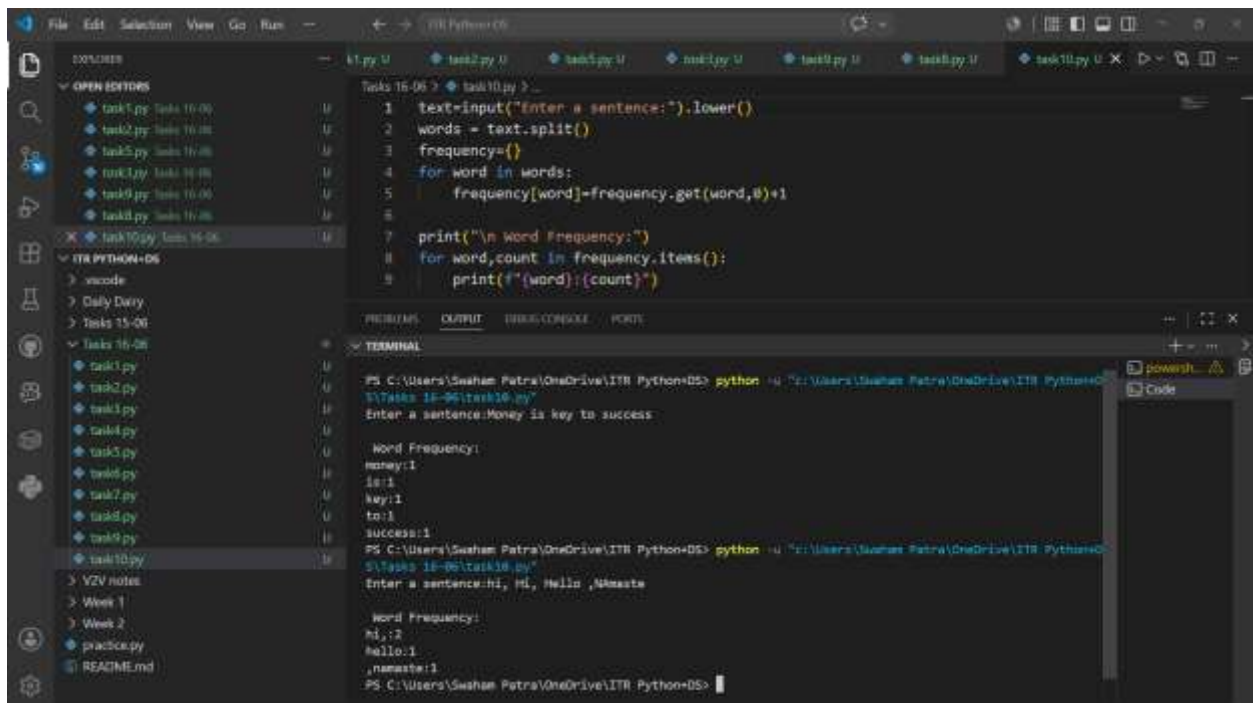
The screenshot shows the VS Code editor with a file explorer on the left containing a directory named 'TASKS' with subdirectories '15-06' and '16-06'. The '16-06' directory contains files 'task1.py' through 'task10.py'. The main editor window displays 'task9.py' with the following code:

```
1 while True:
2     try:
3         num=int(input("Enter the number: "))
4         print("You entered:",num)
5         break
6     except ValueError:
7         print("Invalid input. Try again.")
8
```

The bottom panel shows the 'TERMINAL' output:

```
PS C:\Users\Saaham Patra\OneDrive\ITR Python> python -u "C:\Users\Saaham Patra\OneDrive\ITR Python>
S:\Tasks 16-06\task9.py"
Enter the number: 12
You entered: 12
PS C:\Users\Saaham Patra\OneDrive\ITR Python> python -u "C:\Users\Saaham Patra\OneDrive\ITR Python>
S:\Tasks 16-06\task9.py"
Enter the number: qw
Invalid input. Try again.
Enter the number: 12
You entered: 12
PS C:\Users\Saaham Patra\OneDrive\ITR Python>
```

Task 10:



The screenshot shows the VS Code editor with the same file explorer as Task 9. The main editor window displays 'task10.py' with the following code:

```
1 text=input("Enter a sentence:").lower()
2 words = text.split()
3 frequency={}
4 for word in words:
5     frequency[word]=frequency.get(word,0)+1
6
7 print("\n Word Frequency:")
8 for word,count in frequency.items():
9     print(f"{word}:{count}")
```

The bottom panel shows the 'TERMINAL' output:

```
PS C:\Users\Saaham Patra\OneDrive\ITR Python> python -u "C:\Users\Saaham Patra\OneDrive\ITR Python>
S:\Tasks 16-06\task10.py"
Enter a sentence:Money is key to success

 Word Frequency:
Money:1
is:1
key:1
to:1
success:1
PS C:\Users\Saaham Patra\OneDrive\ITR Python> python -u "C:\Users\Saaham Patra\OneDrive\ITR Python>
S:\Tasks 16-06\task10.py"
Enter a sentence:hi, hi, Hello ,Namaste

 Word Frequency:
hi:2
hello:1
,namaste:1
PS C:\Users\Saaham Patra\OneDrive\ITR Python>
```